

Higher Tier only questions

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Samples A, B and C contain magnesium or calcium ions ☒						
				Only samples A and B contain magnesium or calcium ions ☐			1	1		
				Only samples B and C contain magnesium or calcium ions ☐						
				None of the samples contain magnesium or calcium ions ☐						
	(b)			To show the expected result for permanent hard water ☐						
				To show the expected result for temporary hard water ☐						
				To act as a control for the experiment ☒	1			1		1
				To make sure the investigation is a fair test ☐						
	(c)			volume of soap solution and number of times shaken ☐						
				volume of water sample and volume of soap solution used ☐						
				volume of water sample and number of times shaken ☒	1			1		1
				volume of water sample, volume of soap solution and number of times shaken ☐						
	(d)			Sample B contains temporary hardness only ☐						
				Sample B contains permanent hardness only ☐						
				Sample B contains a mixture of temporary and permanent hardness ☒						
				Sample B does not contain hardness ☐						
				the volume of soap needed after boiling is less than before / less soap is needed after boiling (1)						
				the volume of soap after boiling is still more than the distilled water (1)			3	3		3